Ridge Regression, Gradient Descent, And Perceptron

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Revision of Standard Linear Regression Model

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Problems of Linear Regression

(With Linearly Separable Data)

Problems of Linear Regression

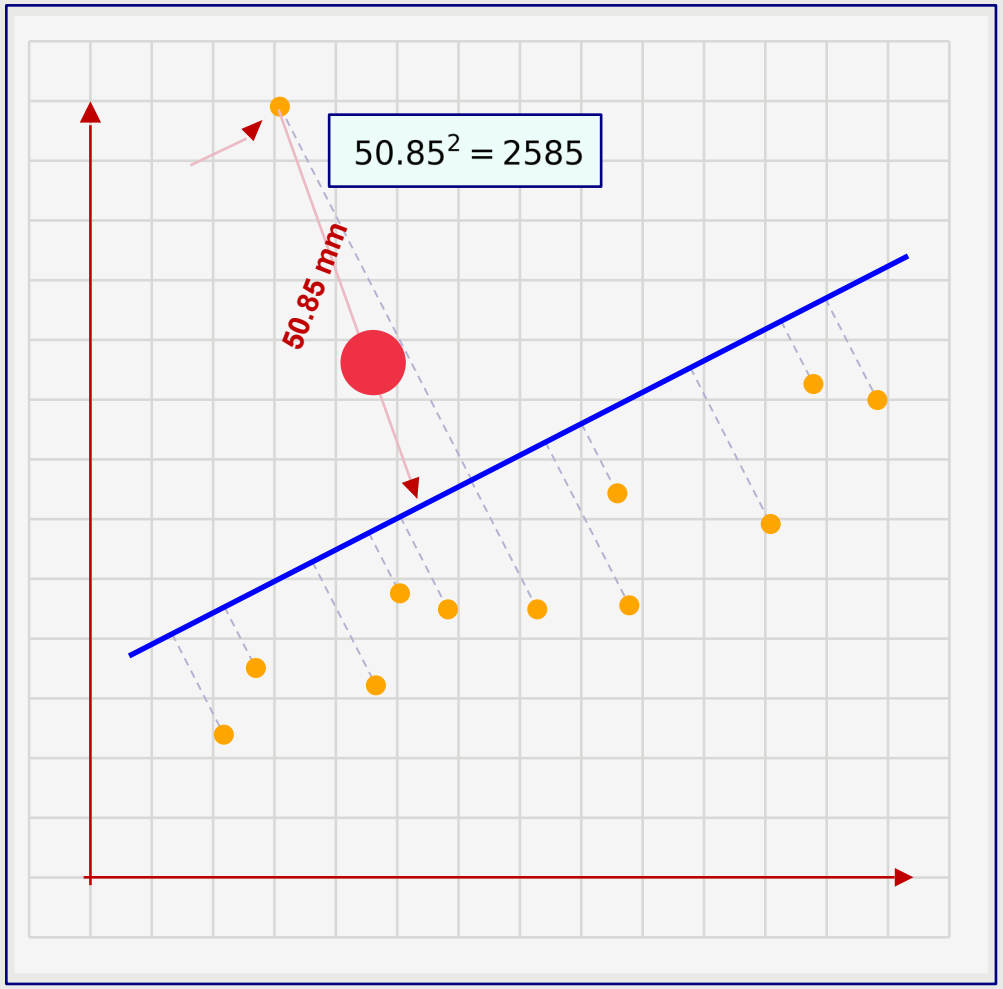
Assume the Dataset is linearly separable (suitable to apply Linear regression model)

Linear regression model has FOUR problems:

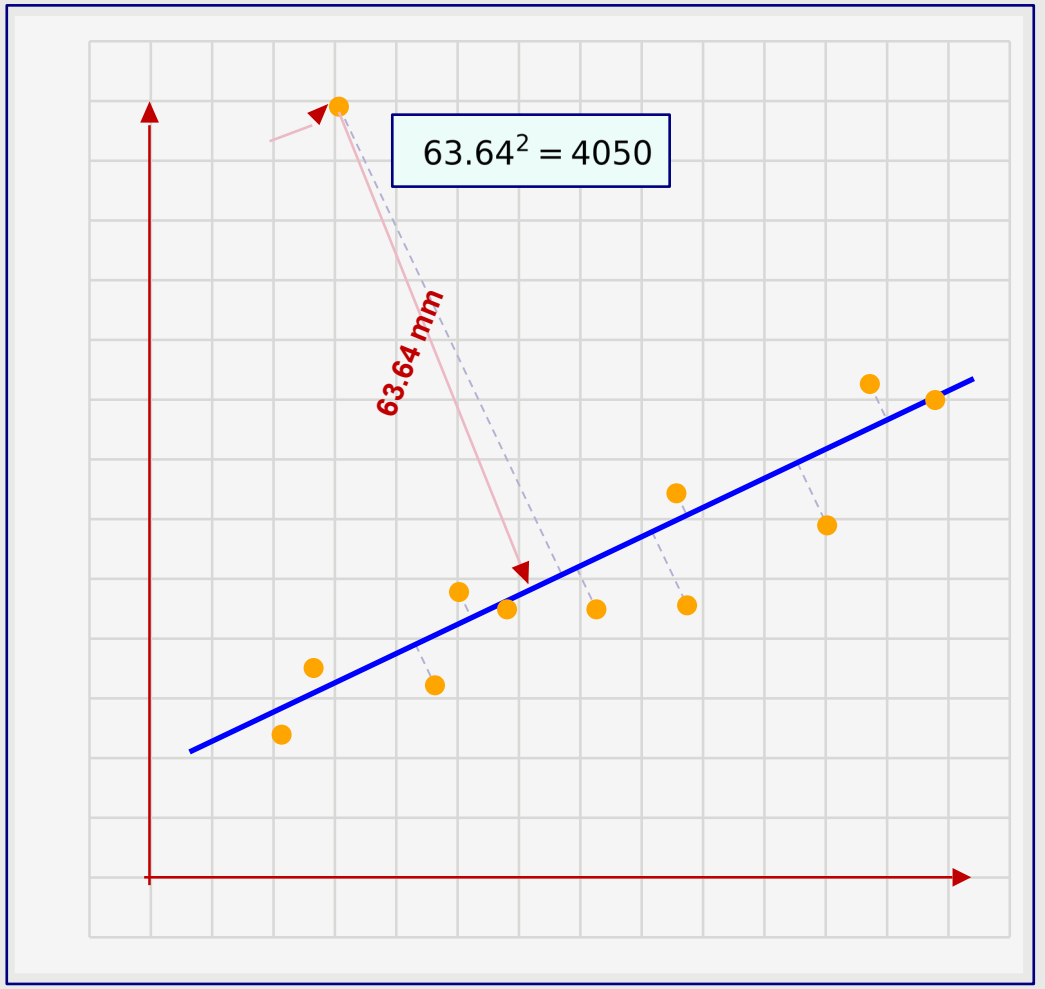
- **Overfitting Problem**
- **Matrix Inversion Problem [1]** ***(Singular Matrix)***
- **Matrix Inversion Problem [2]** ***(Large Matrix dimension)***
- **Different Hyperplanes can solve the problems** ***(Selection of Best Hyperplane)***

[1] Ridge Regularizer to Minimize Overfitting

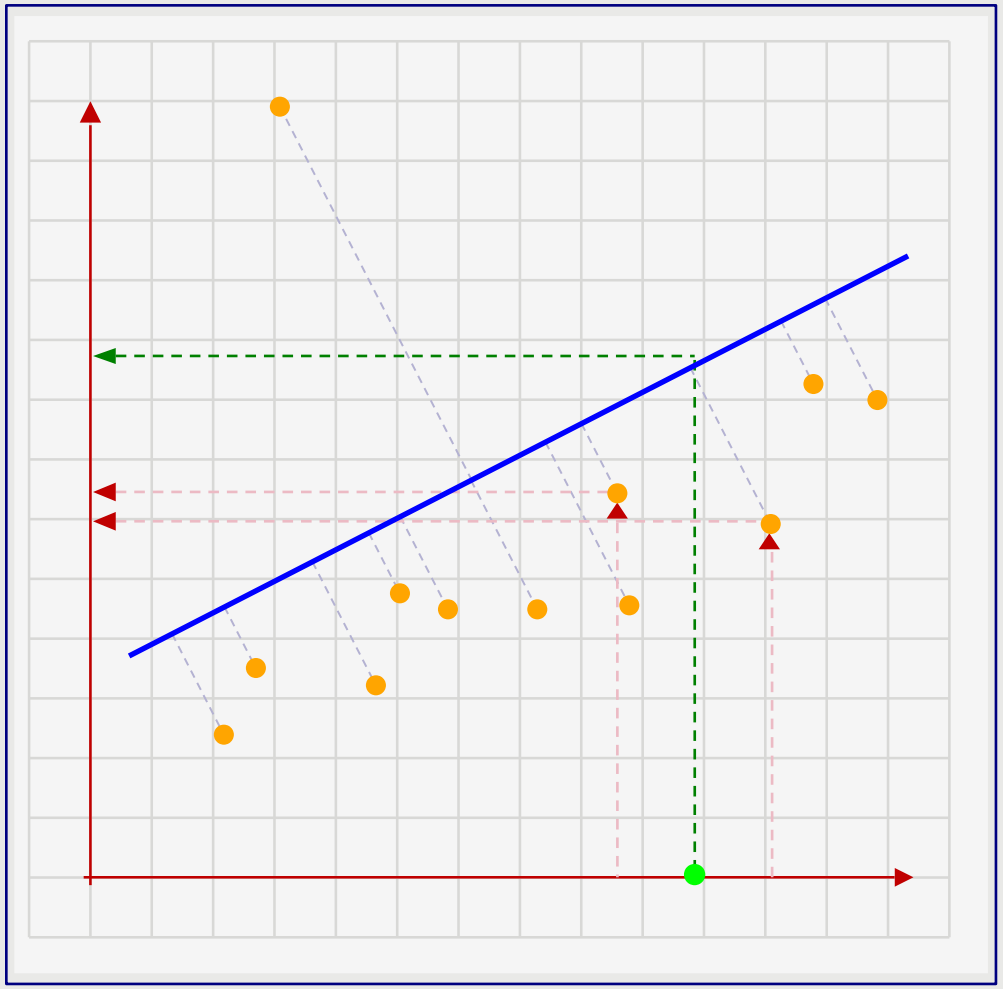
(Also Known as L2 regularizer)



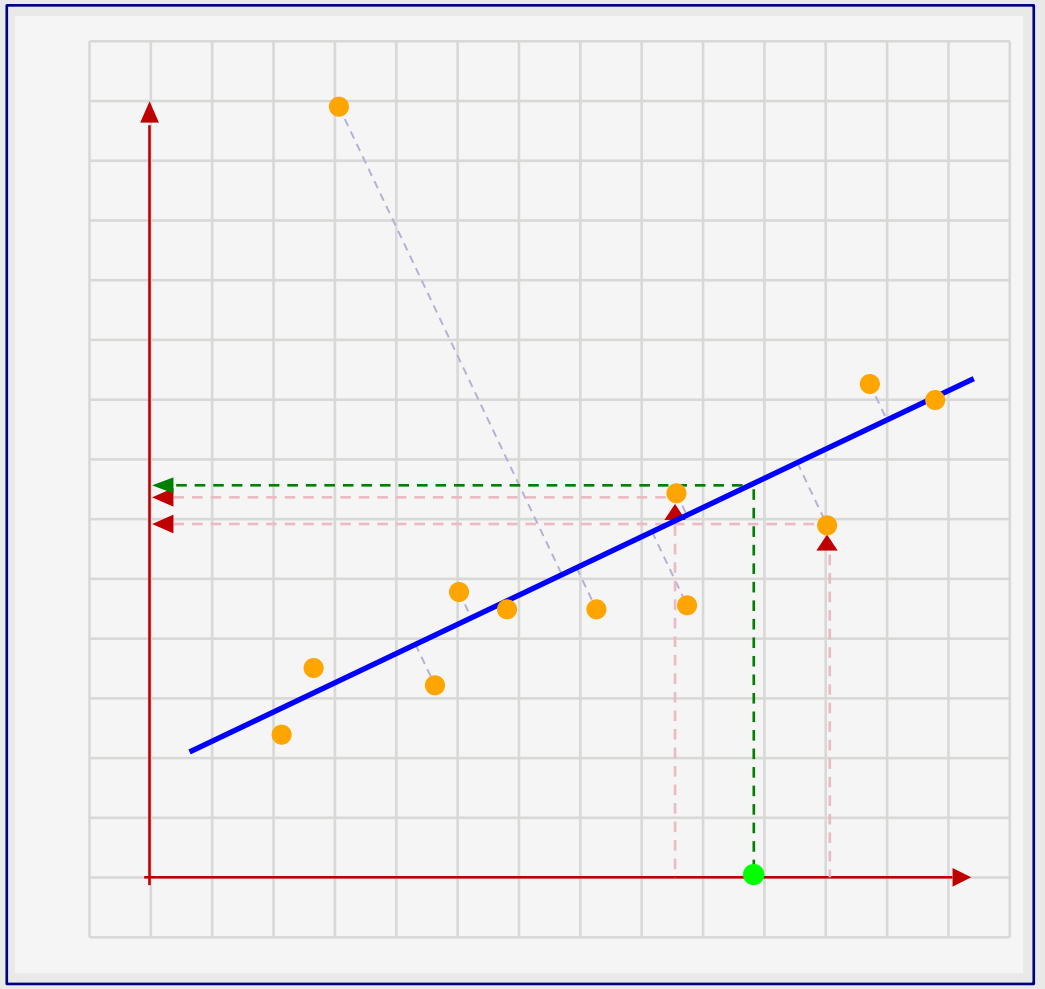
Overfit (try to minimize the residuals ONLY)
Fails to detect the Pattern



Generalized (Residuals are not minimized)
Successes to detect the Pattern



Overfit (try to minimize the residuals ONLY)
Fails to detect the Pattern



Generalized (Residuals are not minimized)
Successes to detect the Pattern

Example of Overfitting due to outlier in data

(and how to minimize overfitting)

Effect of Overfitting on model parameter "W"

Training Data

Perfect Data
Y follows x1

	X1	X2	X3	X4	X5	X6	X7	X8
X1	1	2	3	4	8	9	10	11
X2	1	3	5	6	16	7	5	3
y	2	4	6	8	16	18	20	22

X5 is Outlier

	X1	X2	X3	X4	X5	X6	X7	X8
X1	1	2	3	4	8	9	10	30
X2	1	3	5	6	16	7	5	3
y	2	4	6	8	16	18	20	22

X5 is Outlier

	X1	X2	X3	X4	X5	X6	X7	X8
X1	1	2	3	4	8	9	10	60
X2	1	3	5	6	16	7	5	3
y	2	4	6	8	16	18	20	22

Test Data

	Test1	Test2	Test3
X1	30	40	50
X2	15	20	25
y	60	80	100

Again
Y follows x1

What will be the effect of X8 Outlier values on the trained model ???

Effect of Overfitting on model parameter "W"

Train

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
X_1	1	2	3	4	8	9	10	11
X_2	1	3	5	6	16	7	5	3
y	2	4	6	8	16	18	20	22

$$W = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$$

$$\|W\|^2 = 4$$

Test

$Test_1$	$Test_2$	$Test_3$
30	40	50
15	20	25
60	80	100

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
X_1	1	2	3	4	8	9	10	30
X_2	1	3	5	6	16	7	5	3
y	2	4	6	8	16	18	20	22

$$W = \begin{pmatrix} 2.7 \\ 0.67 \\ 0.64 \end{pmatrix}$$

$$\|W\|^2 = 8.11$$

$Test_1$	$Test_2$	$Test_3$
30	40	50
15	20	25
32.4	42.3	52.3

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
X_1	1	2	3	4	8	9	10	60
X_2	1	3	5	6	16	7	5	3
y	2	4	6	8	16	18	20	22

$$W = \begin{pmatrix} 4 \\ 0.29 \\ 0.78 \end{pmatrix}$$

$$\|W\|^2 = 16.7$$

$Test_1$	$Test_2$	$Test_3$
30	40	50
15	20	25
24.4	31.2	38

pending recreation

pending recreation

L_1 and L_2 Regularization

Regularizers are based on:

Penalize large values of W (model parameters)

- This have the effect of better generalization of the model
- Since there is a contradiction between model training (using training dataset) and model generalization, the generalization term is added to the objective loss function with a customized weigh factor (do regularization to what extent)
- Penalization may be applied on the Squared Norm of the Weight Vector
  This is called L_2 regularization (also know as Ridge Regularization)
- Penalization may also be applied on the sum of Absolute values of the Weight Vector
  This is called L_1 regularization
- Regularization is particularly important for Small number of samples (N) and Large vector Length (d)

Adding Ridge Regularizer to the model λ=3

Train

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
X_1	1	2	3	4	8	9	10	11
X_2	1	3	5	6	16	7	5	3
y	2	4	6	8	16	18	20	22

$$W = \begin{pmatrix} 0.1 \\ 1.95 \\ 0.03 \end{pmatrix}$$

$$\|W\|^2 = 3.8$$

Test

	$Test_1$	$Test_2$	$Test_3$
X_1	30	40	50
X_2	15	20	25
y	59	78.7	98.4

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
X_1	1	2	3	4	8	9	10	30
X_2	1	3	5	6	16	7	5	3
y	2	4	6	8	16	18	20	22

$$W = \begin{pmatrix} 1.2 \\ 0.72 \\ 0.77 \end{pmatrix}$$

$$\|W\|^2 = 2.45$$

	$Test_1$	$Test_2$	$Test_3$
X_1	30	40	50
X_2	15	20	25
y	34.1	45.1	56.1

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
X_1	1	2	3	4	8	9	10	60
X_2	1	3	5	6	16	7	5	3
y	2	4	6	8	16	18	20	22

$$W = \begin{pmatrix} 1.77 \\ 0.32 \\ 0.99 \end{pmatrix}$$

$$\|W\|^2 = 4.2$$

	$Test_1$	$Test_2$	$Test_3$
X_1	30	40	50
X_2	15	20	25
y	26.2	34.4	42.5

Adding Ridge Regularizer to the model λ=10

Train

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
X_1	1	2	3	4	8	9	10	11
X_2	1	3	5	6	16	7	5	3
y	2	4	6	8	16	18	20	22

$$\underline{W} = \begin{vmatrix} 0.13 \\ 1.87 \\ 0.09 \end{vmatrix}$$

$$\|W\|^2 = 3.5$$

Test

	$Test_1$	$Test_2$	$Test_3$
X_1	30	40	50
X_2	15	20	25
y	57.4	76.5	95.6

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
X_1	1	2	3	4	8	9	10	30
X_2	1	3	5	6	16	7	5	3
y	2	4	30	8	16	18	20	22

$$\underline{W} = \begin{vmatrix} 0.54 \\ 0.73 \\ 0.81 \end{vmatrix}$$

$$\|W\|^2 = 1.48$$

	$Test_1$	$Test_2$	$Test_3$
X_1	30	40	50
X_2	15	20	25
y	34.6	46.4	57.4

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
X_1	1	2	3	4	8	9	10	60
X_2	1	3	5	6	16	7	5	3
y	2	4	60	8	16	18	20	22

$$\underline{W} = \begin{vmatrix} 0.82 \\ 0.33 \\ 1.1 \end{vmatrix}$$

$$\|W\|^2 = 1.92$$

	$Test_1$	$Test_2$	$Test_3$
X_1	30	40	50
X_2	15	20	25
y	26.8	35.4	44.1

pending recreation

[2] Ridge Regularization to overcome

Inverse Problem of Singular Matrix

Inverse Problem of Singular Matrix

$X^T X$

14	10	15
10	40	46
20	80	92

$|X^T X| = 0$ ⇒ Singular Matrix ⇒ *No Inverse*

$(X^T X + \lambda I)$ $\lambda = 3$

17	10	15
10	43	46
20	80	95

$|X^T X| = 5685$ ⇒ Non-Singular Matrix ⇒ *Inverse Exists*

$(X^T X + \lambda I)^{-1}$ ⇒

0.07	0.04	-0.03
-0.01	0.23	-0.11
-0.01	-0.2	0.11

[3] Inverse of High Dimension (non-singular) Matrix

Inverse of High Dimension Matrix

Recall

$$\underline{W} = (X^T X + \lambda I)^{-1} X^T \underline{Y}$$

if Feature vectors are of dimension d $\Rightarrow$ ($\underline{X}^T X$) is of size d+1 x d+1

The standard matrix inversion algorithm has a time complexity of $O(n^3)$, which means that the computation time increases rapidly with the size of the matrix (increasing cubically with the number of components).

For Example: if Features vector is of size 1000 then Inverse calculation is or order 1000,000,000 which is impractical

Solution:

*Instead of using **Least Squares** with Matrix inversion,*
*use **Gradient Descent** iterative solution*

Gradient Descent (GD)

and

How to apply to minimize Loss function

pending recreation

Gradient Descent Solution

Gradient Descent (GD) Algorithm:

[1] Start with initial **Random Weights (W)**

[2] Update the weights by moving along the direction of **negative gradient** of Loss function (w.r.t. W) (**moving step is a hyper parameter, AKA Learning Rate**)

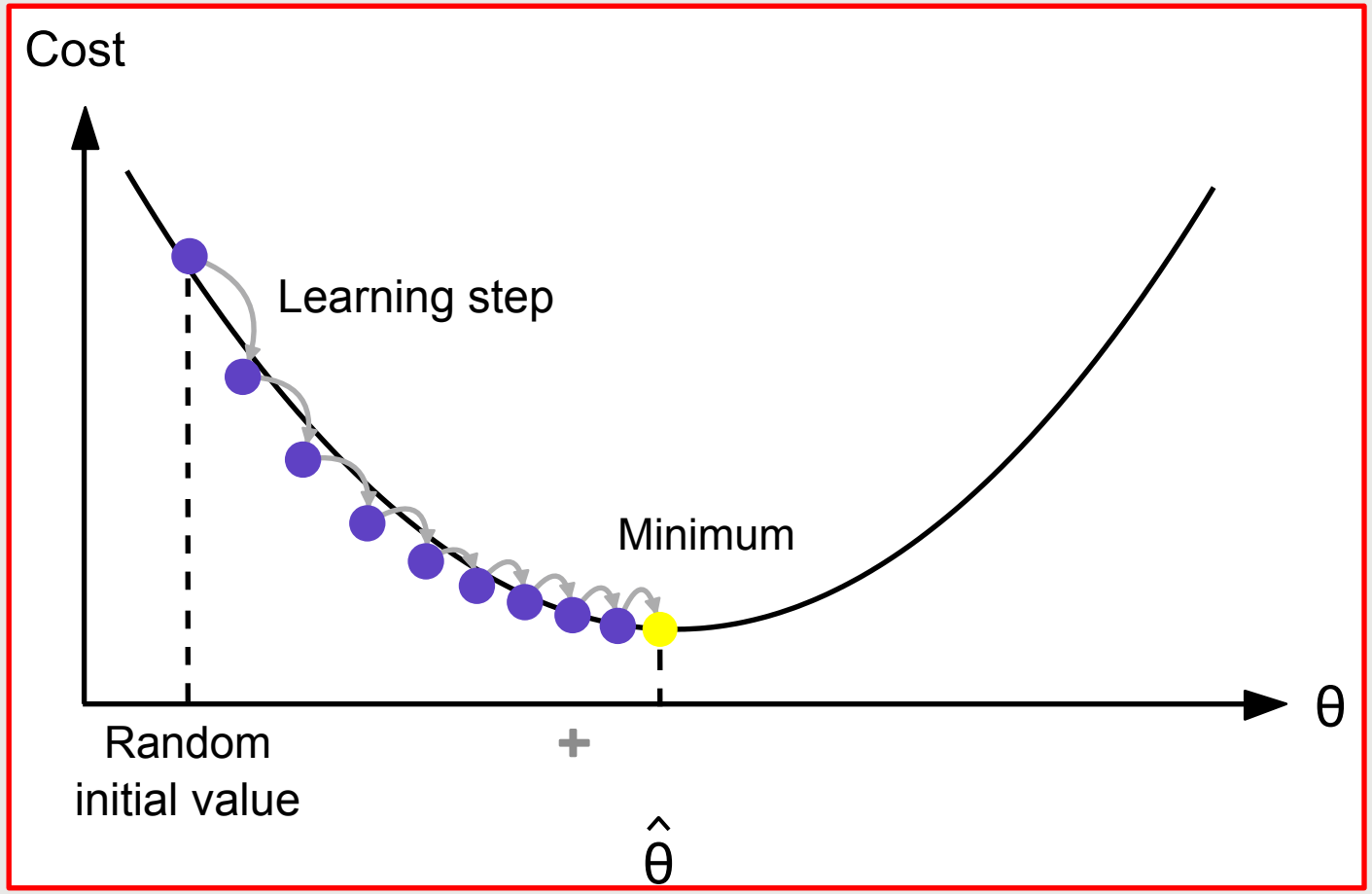
[3] Stop when the gradient is **almost zero** (this is at the minimum value of the function)

Notes:

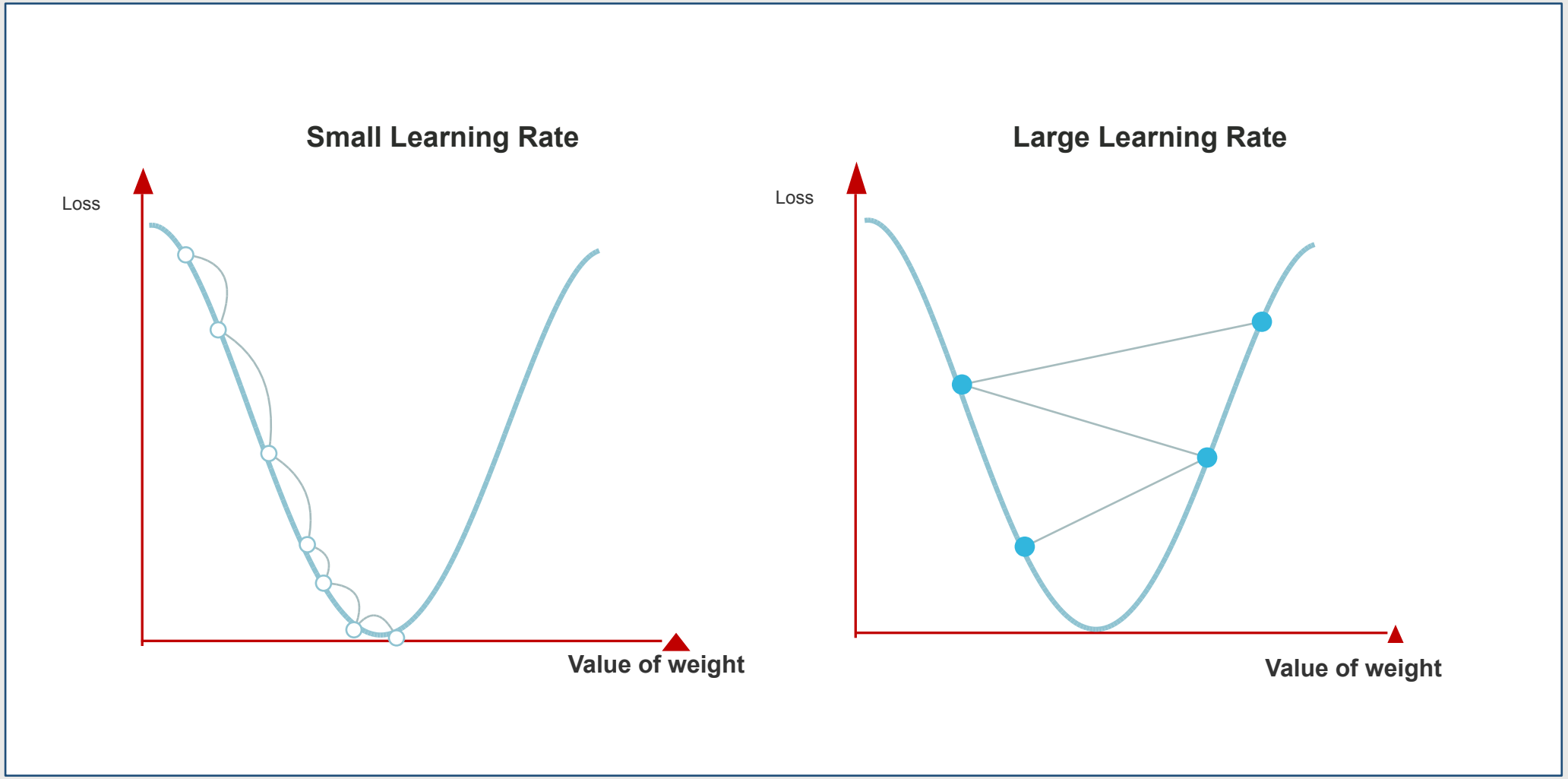
- **Least Squares** Linear regression objective function **is a convex function** (even with regularizer). This means it has **ONE Minimum** (good news).

- GD will approximatly find this minimum value but at a variable number of steps depending on the used **Learning Rate**

Follow Negative Direction of Gradient with Selected Learning Rate

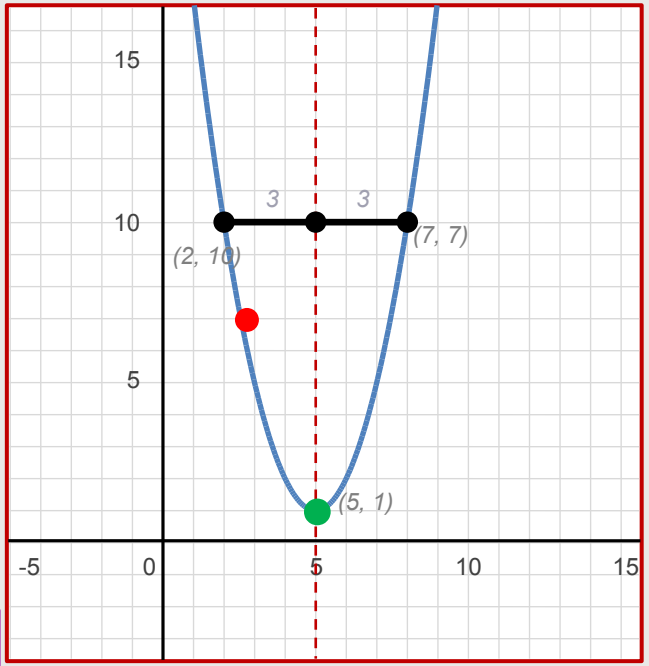


Effect of small and large Learning Rate



Example: Obtain Minimum Value of a function using Gradient Descent (GD)

Convex Function, Learning Rate =0.2, Start at X=3



step	0.2
initial X	3
delta	0.1

$f(x) = (x-5)^2 + 1$

$X = 5$

f(x) Minimum value is **1.0**

Reach Minimum after 11 Steps

x	3	3.2	3.4	3.6	3.8	4	4.2	4.4	4.6	4.8	5
f(x)	5	4.24	3.56	2.96	2.44	2	1.64	1.36	1.16	1.04	1
x-delta	2.9	3.1	3.3	3.5	3.7	3.9	4.1	4.3	4.5	4.7	4.9
f(x-delta)	5.41	4.61	3.89	3.25	2.69	2.21	1.81	1.49	1.25	1.09	1.01
x+delta	3.1	3.3	3.5	3.7	3.9	4.1	4.3	4.5	4.7	4.9	5.1
f(x+delta)	4.61	3.89	3.25	2.69	2.21	1.81	1.49	1.25	1.09	1.01	1.01
slope	-8	-7.2	-6.4	-5.6	-4.8	-4	-3.2	-2.4	-1.6	-0.8	0
next x	3.2	3.4	3.6	3.8	4	4.2	4.4	4.6	4.8	5	stop

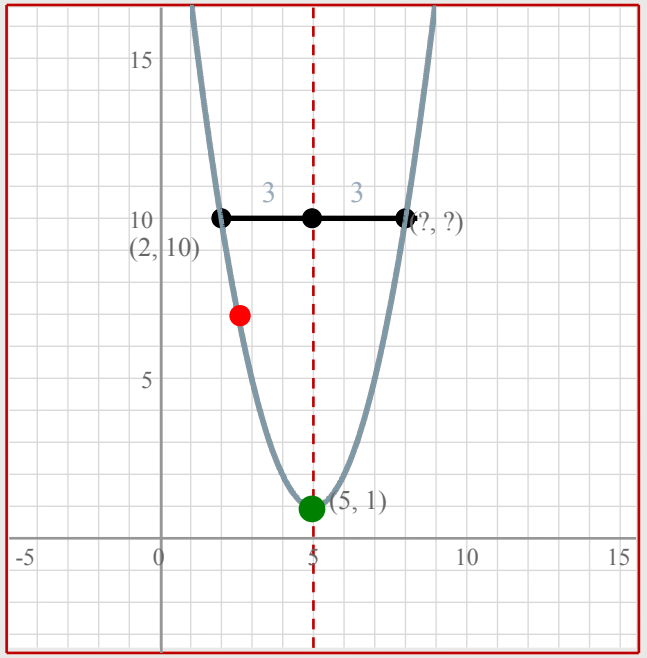
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Convex Function, Learning Rate =0.4, Start at X=3

step	0.4
initial X	3
delta	0.1

$$f(x) = (x-5)^2 + 1$$

***X* = 5**
f(x) Minimum value is **1.0**



Reach Minimum after 6 Steps

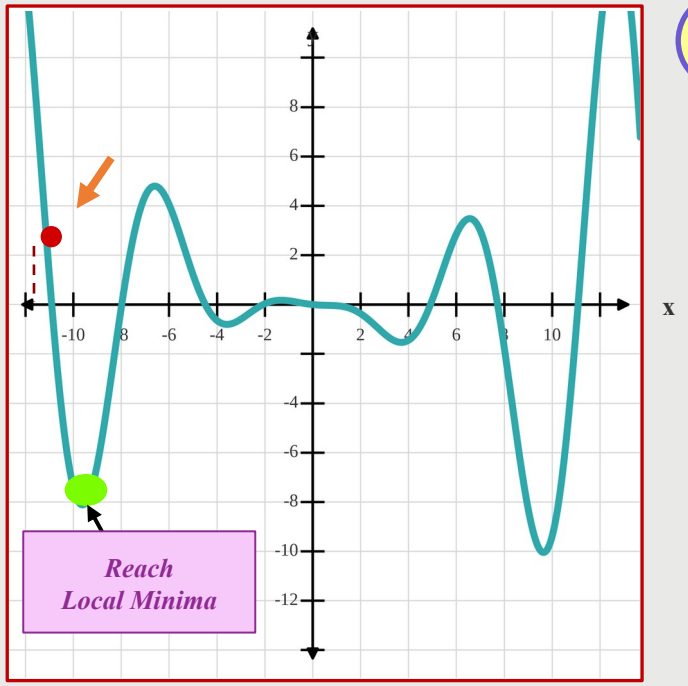
x	3	3.4	3.8	4.2	4.6	5
f(x)	5	3.56	2.44	1.64	1.16	1
x-delta	2.9	3.3	3.7	4.1	4.5	4.9
f(x-delta)	5.41	3.89	2.69	1.81	1.25	1.01
x+delta	3.1	3.5	3.9	4.3	4.7	5.1
f(x+delta)	4.61	3.25	2.21	1.49	1.09	1.01
slope	-8	-6.4	-4.8	-3.2	-1.6	0
next x	3.4	3.8	4.2	4.6	5	stop

Non-Convex Function

May reach different minimum value based on:

initial point and learning rate

$$f(x) = \frac{x^2\cos(x) - x}{10}$$



step	0.1
initial X	-11
delta	0.1

X is in between -9.7 and color red bold -9.6

f(x) is in between -8.08 and -8.011

x	-11	-10.9	-10.8	-10.7	-10.6	-10.5	-10.4	-10.3	-10.2	-10.1	-10	-9.9	-9.8	-9.7	-9.6	-9.7	-9.6	-9.7
f(x)	1.15	-0.04	-1.19	-2.26	-3.27	-4.19	-5.03	-5.77	-6.41	-6.95	-7.39	-7.72	-7.96	-8.08	-8.11	-8.08	-8.11	-8.08
x-delta	-11.1	-11	-10.9	-10.8	-10.7	-10.6	-10.5	-10.4	-10.3	-10.2	-10.1	-10	-9.9	-9.8	-9.7	-9.8	-9.7	-9.8
f(x-delta)	2.39	1.15	-0.04	-1.19	-2.26	-3.27	-4.19	-5.03	-5.77	-6.41	-6.95	-7.39	-7.72	-7.96	-8.08	-7.96	-8.08	-7.96
x+delta	-10.9	-10.8	-10.7	-10.6	-10.5	-10.4	-10.3	-10.2	-10.1	-10	-9.9	-9.8	-9.7	-9.6	-9.5	-9.6	-9.5	-9.6
f(x+delta)	-0.04	-1.19	-2.26	-3.27	-4.19	-5.03	-5.77	-6.41	-6.95	-7.39	-7.72	-7.96	-8.08	-8.11	-8.05	-8.11	-8.05	-8.11
slope	-24.3	-23.4	-22.2	-20.8	-19.3	-17.6	-15.8	-13.8	-11.8	-9.8	-7.7	-5.7	-3.6	-1.5	0.3	-1.5	0.3	-1.5
next x	-10.9	-10.8	-10.7	-10.6	-10.5	-10.4	-10.3	-10.2	-10.1	-10	-9.9	-9.8	-9.7	-9.6	-9.7	-9.6	-9.7	-9.6

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